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Marketing Capstone Project - MKT 7970

Project Title: Predictive Analysis of Restaurant Revenue

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1. Executive Summary

This project analyzes the key variables influencing restaurant revenue and explores how data can assist managers in optimizing daily operations, staffing, pricing, and marketing strategies. The dataset includes hundreds of restaurant observations that include important business information like menu prices, seating capacity, ratings and reviews, chef experience, ambiance scores, reservations, marketing budgets, and types of cuisine. Upon cleaning and organizing the data, I used descriptive analytics, regression modeling, and machine learning methodologies in Python, afterwards visualizing the main insights in Tableau.

The descriptive results showed clear patterns: restaurants with higher menu prices and more seating always make higher profits. Tableau also showed big differences between cuisines, with Japanese and French restaurants making the most money. These patterns helped us model.

I used two prediction models to find out which factors have the biggest effect on revenue: Linear Regression to help us understand and Random Forest to make sure our predictions were right. It did a great job; it got a R^2 of 0.999, which means it caught almost all of the changes in income. Linear Regression also did well ($R^2 = 0.955$), and its p-values and coefficients showed that the number of seats, the average price of a meal, and the experience of the chef were all good indicators.

The Random Forest model confirmed these findings, identifying Average Meal Price and Seating Capacity as the main predictors. The scatter plot showed an almost perfect diagonal line, indicating extremely precise revenue forecasts. According to the model outcomes and visual evaluation, various practical approaches were proposed for restaurant managers:

Key Strategic Implications

- **Pricing:** Use premium pricing when it makes logic, add high-value items, and make sure prices are in line with quality.
- **Operations (Seating):** Add more seats where you can, change the layout to make it easier for people to get in and out, and use reservation systems to make the most of your space.
- **Talent (Chef Experience):** Hire and keep chefs with a lot of experience, give them training, and use marketing to show off their skills.
- **Positioning Cuisine:** Market cuisines that do well (like Japanese and French) and change menus to meet demand.

Overall, the experiment shows that the price of food, the number of seats, and the chef's experience have the most effects on restaurant sales. The type of food and the ratings have less of an effect. This study employs Python and Tableau to provide clear, data-driven insights that managers may use to improve pricing, operations, and overall profitability.

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2. The Problem

Restaurants today face uncertainty about what truly drives revenue. Many businesses invest in social media ads, promotions, loyalty rewards, and ambiance upgrades but these strategies do not always lead to higher revenue without clear insights, managers struggle to make informed decisions.

The objective of this project is to determine the key predictors of restaurant revenue and to develop precise predictive models utilizing real-world variables such as pricing, reviews, reservations, cuisine type, marketing expenditure, and customer experience indicators.

Understanding these drivers is crucial because:

- Restaurants must optimize pricing without losing customers
- Marketing budgets are limited
- Competition is increasing
- Consumer behavior fluctuates
- Operational efficiency impacts table turnover and revenue

This project offers data-driven insights into the most significant factors.

3. The Data

The dataset includes thousands of restaurant observations containing variables such as:

- Revenue
- Seating capacity
- Average meal price
- Chef experience years
- Ratings and reviews
- Social media engagement
- Marketing budget
- Reservation data
- Cuisine types (encoded as dummy variables)

3.1 Descriptive Analysis

Four key visualizations were created

a. Revenue Histogram:

The revenue histogram provides a clear overview of the distribution of restaurant profits within the dataset. Each bar indicates a revenue range (bin), and the height of the bar reflects the number of restaurants within that category. The chart shows multiple significant patterns:

- Most restaurants fall in the mid-revenue range, especially between \$450K and \$800K which means most restaurants earn moderate but stable revenue.
- Revenue peaks around the \$550K–\$650K range, where the bars are tallest which shows that many restaurants consistently cluster around this earning level.
- Very low-revenue and very high-revenue restaurants are less common, as shown by the shorter bars on both ends of the chart.
- The histogram also shows a gradual decline after the peak, indicating that while some restaurants earn over \$1 million, the numbers become fewer as revenue increases.

Overall, this histogram helps us understand the earning pattern across restaurants. Most perform in the middle range, while only a small portion reach very high or very low revenue levels.

Distribution of Restaurant Revenue

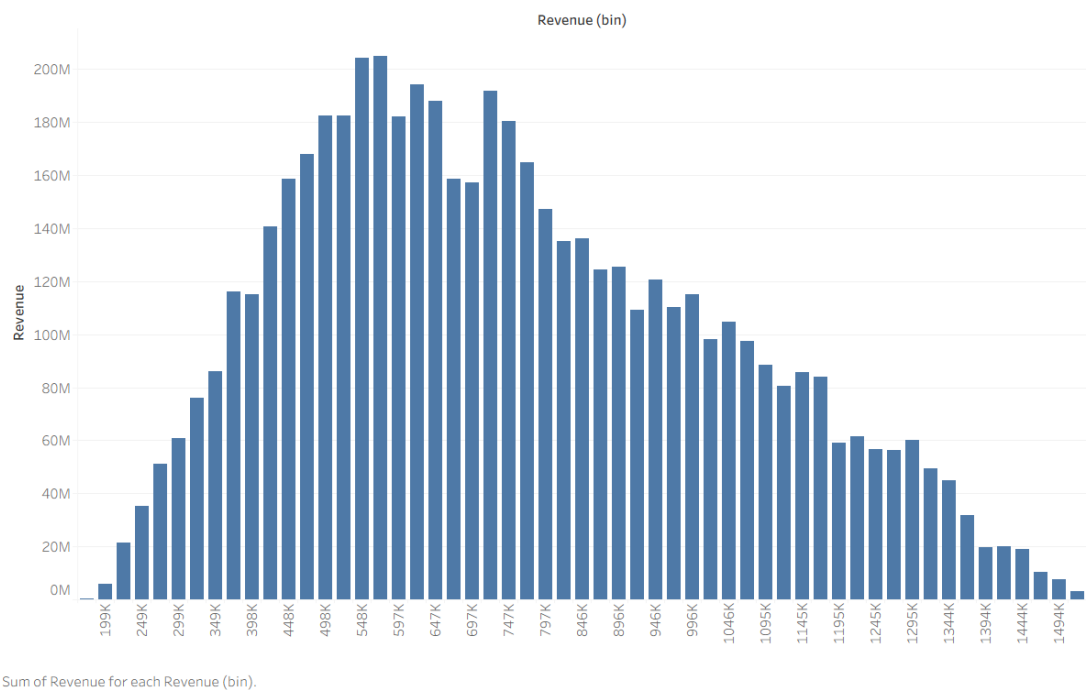
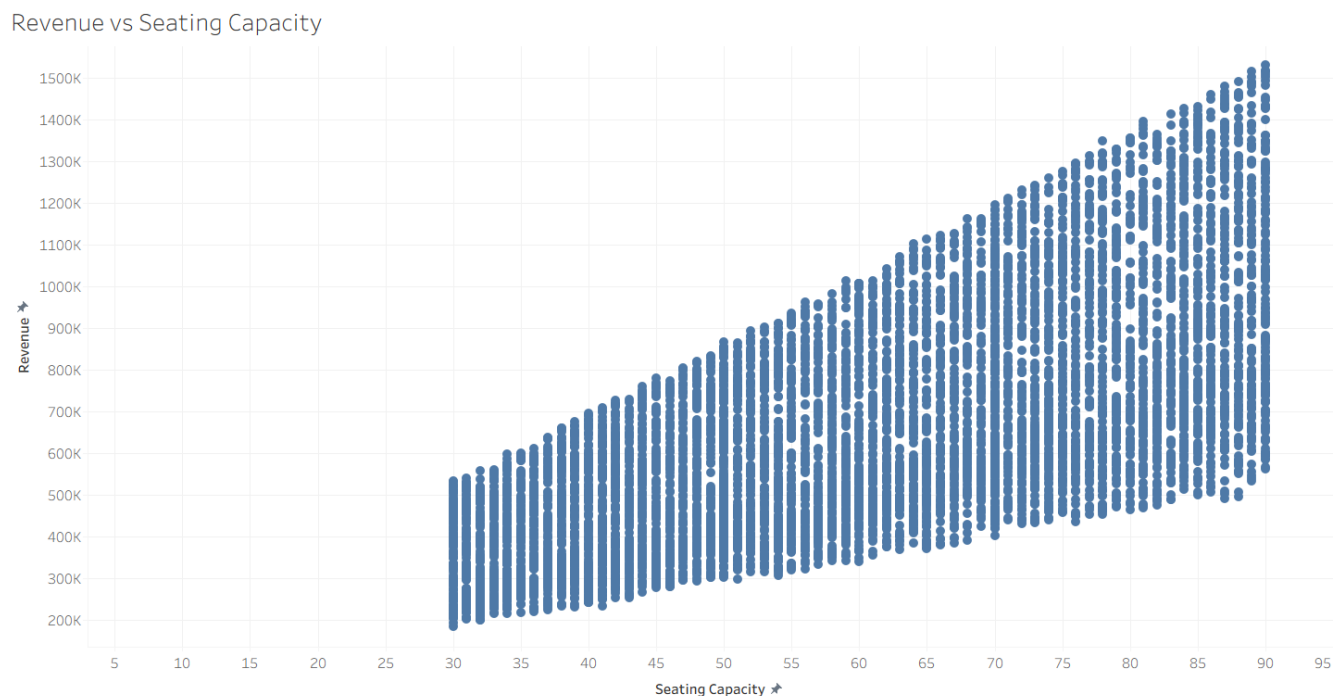


Figure 1: Distribution of Restaurant Revenue

b. Seating Capacity vs Revenue

This scatter plot shows the correlation between a restaurant's seating capacity and its overall revenue. Each point matches a restaurant, placed according to its seating capacity and revenue generated. Multiple significant patterns appear:

- **Clear upward trend:** As the seating capacity increases, revenue correspondingly increases. Restaurants that have more seating capacity can accommodate a higher number of customers at the same time, thereby simply increasing revenue.
- **Strong positive correlation:** The data points create an upward "band," indicating that seating capacity is one of the most important variables affecting revenue within the dataset.
- **Wider spread at higher capacities:** For restaurants with very high seating (80–95 seats), revenue varies more widely. This shows that while more seats generally increase revenue, other factors (like pricing, chef experience, or service quality) still play a role at higher scales.
- **Lower-capacity restaurants earn significantly less:** Restaurants with around 30–40 seats consistently stay in the lower revenue range, reinforcing the impact of limited capacity.



Seating Capacity vs. Revenue.

Figure 2: Revenue vs Seating Capacity

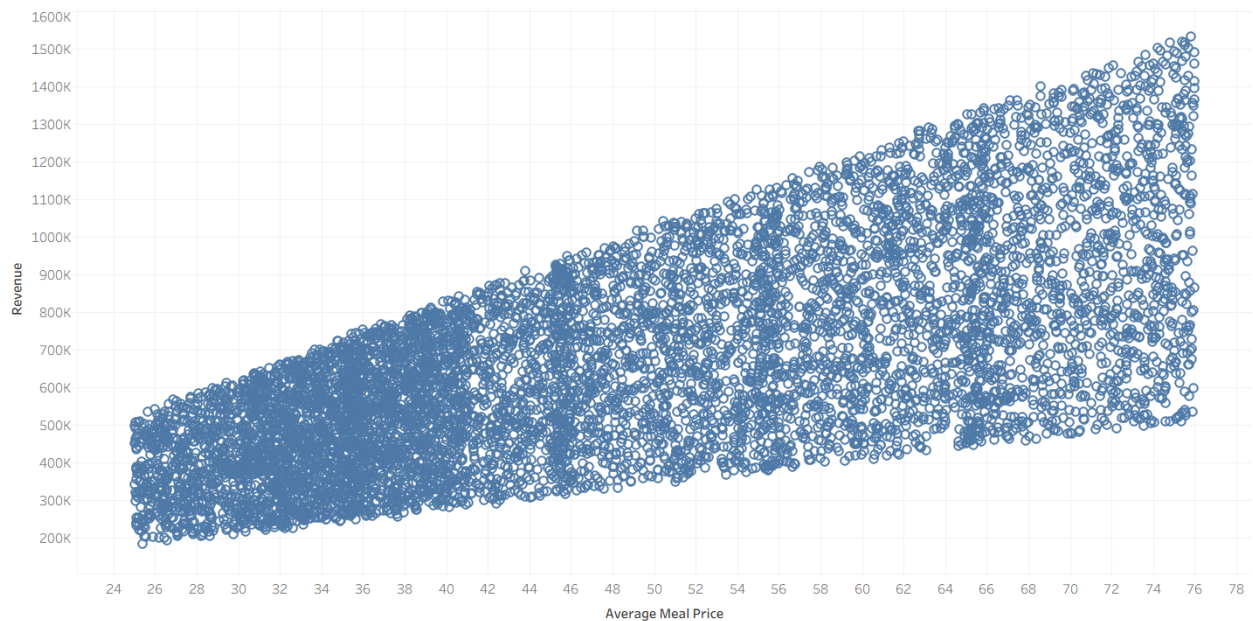
c. Average Meal Price vs Revenue

This scatter plot shows the strong and consistent correlation between the average meal price charged by a restaurant and its total revenue. As the average cost of a meal increases, the revenue points demonstrate a clearly upward trend.

Although there is some fluctuation in the data due to differences in restaurant sizes, cuisines, and customer demographics, the overall pattern remains clearly visible. Restaurants with increased menu prices usually generate significantly more revenue.

It shows from this that pricing is a key element in the process of generating revenue. The ability to demand higher prices allows restaurants that serve higher-value or premium food to improve their revenue performance by allowing them to cover their higher costs. In addition, it highlights the fact that customers are frequently willing to pay a higher price when they believe that the quality of the meal, the service, or the overall experience is worth the price.

Revenue vs Average Meal Price



Average Meal Price vs. Revenue.

Figure 3: Revenue vs Average Meal Price

d. Average Revenue by Cuisine Type

This bar chart illustrates the average revenue produced by various cuisine types, with the differences being very clear. Japanese restaurants perform very well, with an average income of over \$937,000, which is the most of any type of restaurant. This shows that Japanese food is in high demand and that people are willing to pay more for it. Then there is French cuisine, which is also growing quickly and making about \$820,000 a year. Next are Italian eateries, which make a lot of money: \$692,000. Other types of food, like Indian, Mexican, and "Other," make less money on average, between \$427,000 and \$564,000. This does not mean that these restaurants are not making revenue; it only means that certain types of food cost more, attract different types of customers, or operate in markets with high costs.

Average Revenue by Cuisine

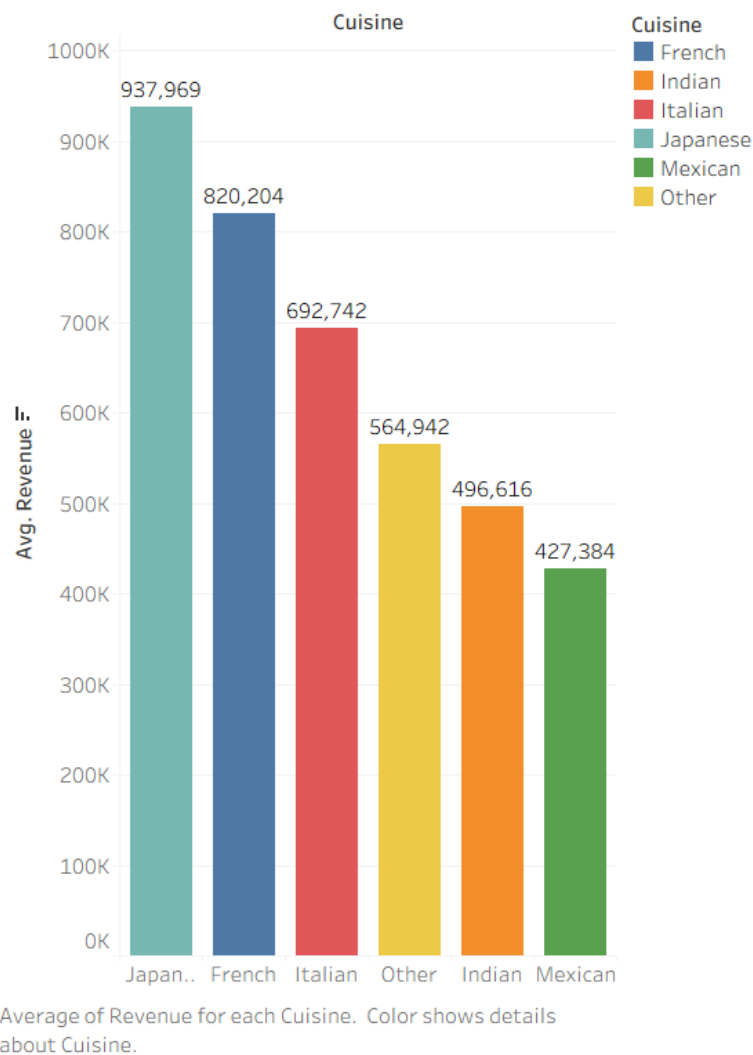


Figure 4: Average Revenue by Cuisine

4. Methodology

4.1 Data Cleaning & Preparation

This table shows a snapshot of the final cleaned dataset after all preprocessing steps were completed. Several important transformations were made:

- **Categorical variables were converted into dummy variables:** Cuisine types such as Japanese, Mexican, Italian, Indian, and French were converted into numerical variables that act as indicators.
- **All numerical variables were cleaned and standardized:** Variables such as seating capacity, average meal price, chef experience, ratings, reviews, marketing budget, and reservations all display clear and consistent values.
- **No missing values remain in the dataset:** The summary on the right shows that each column contains no missing entries, indicating that the data is complete and suitable for modeling.
- **Total Reservations was created as a new feature:** Total Reservations gathers weekday and weekend bookings to provide an increased understanding of customer activity.
- **All variables now have appropriate data types:** All columns are kept in the right format, either as integers or floats.

dummy_location dummy_Japanese dummy_Mexican dummy_Italian dummy_Indian \						[3 rows x 21 columns]	
0	1	1	0	0	0	dummy_location	int64
1	0	0	1	0	0	dummy_Japanese	int64
2	1	0	0	1	0	dummy_Mexican	int64
						dummy_Italian	int64
						dummy_Indian	int64
						dummy_French	int64
dummy_French Rating Seating_Capacity Average_Meal_Price \							
0	0	4.0	38	73.98		Rating	float64
1	0	3.2	76	28.11		Seating_Capacity	int64
2	0	4.7	48	48.29		Average_Meal_Price	float64
						Marketing_Budget	int64
Marketing_Budget ... Chef_Experience_Years Number_of_Reviews \						Social_Media_Followers	int64
0	2224	...	13	185		Chef_Experience_Years	int64
1	4416	...	8	533		Number_of_Reviews	int64
2	2796	...	18	853		Avg_Review_Length	float64
						Ambience_Score	float64
						Service_Quality_Score	float64
Avg_Review_Length Ambience_Score Service_Quality_Score \						dummy_Parking_Availability	int64
0	161.924906	1.3	7.0			Weekend_Reservations	int64
1	148.759717	2.6	3.4			Weekday_Reservations	int64
2	56.849189	5.3	6.7			Total_Reservations	int64
						Revenue	float64
dummy_Parking_Availability Weekend_Reservations Weekday_Reservations \						dtype: object	
0		1	13	4		Missing per column:	
1		1	48	6		dummy_location	0
2		0	27	14		Chef_Experience_Years	0
						Total_Reservations	0
						Weekday_Reservations	0
						Weekend_Reservations	0
						dummy_Parking_Availability	0
						Service_Quality_Score	0
						Ambience_Score	0
						Avg_Review_Length	0
						Number_of_Reviews	0
						dtype: int64	
Total_Reservations Revenue							
0	17	638945.52					
1	54	490207.83					
2	41	541368.62					

Figure 5: Cleaned and encoded dataset after preprocessing

Figure 5: Cleaned and encoded dataset after preprocessing

4.2 Variable Focus

Although the dataset contained many factors, this project focused on those that were most relevant both statistically and from a business perspective. These variables consistently showed significant patterns in the descriptive analysis and often have to do with restaurant performance:

- **Seating Capacity:** Larger restaurants can serve more customers, so this is naturally tied to higher revenue.
- **Average Meal Price:** Pricing directly affects how much money a restaurant earns per customer.
- **Chef Experience:** More experienced chefs often deliver better quality, which can attract more customers and justify higher prices.
- **Cuisine Type:** Some cuisines, like Japanese and French, showed significantly higher average revenue.
- **Ratings & Reviews:** Customer feedback reflects satisfaction and can drive new and repeat visits.
- **Marketing Budget:** Higher marketing investment may increase visibility and customer traffic.

These variables were selected due to their different trends observed in the visualizations, their consistency with restaurant business principles, and their significant connections with revenue. Together, they establish an effective foundation for developing precise predictive models.

4.3 Modeling Approach

Two models were developed: Linear Regression and Random Forest

Linear Regression (Interpretability Model)	Random Forest Regression (Prediction Accuracy Model)
Explain which factors significantly affect revenue	Predict revenue as accurately as possible
Assumes linear relationships between variables	No linear assumptions; captures complex, nonlinear patterns
Very high - provides coefficients, p-values, and effect sizes	Lower - provides feature importance but not coefficients
Coefficients showing direction & strength of each variable	Importance scores showing which features contribute most

Easy to interpret, identifies significant variables, good for business insights	Very accurate predictions, handles nonlinearity, robust to noise & outliers
Struggles with nonlinear relationships; affected by multicollinearity	Harder to interpret; may require more computation
Understanding <i>why</i> revenue changes	Predicting <i>how much</i> revenue a restaurant will earn
$R^2 = 0.955$ (strong)	$R^2 = \mathbf{0.999}$ (excellent)
Identified important predictors: seating, price, chef experience	Confirmed importance ranking; gave highest prediction accuracy

	feature	importance
7	Seating_Capacity	0.504865
8	Average_Meal_Price	0.492786
0	dummy_location	0.000654
11	Chef_Experience_Years	0.000231
13	Avg_Review_Length	0.000127
14	Ambience_Score	0.000127
12	Number_of_Reviews	0.000125
1	dummy_Japanese	0.000125
15	Service_Quality_Score	0.000124
9	Marketing_Budget	0.000119

Figure 6: Top Predictors of Restaurant Revenue (Random Forest)

The figure shows the Random Forest model's most important variables for predicting restaurant income. Each feature gets an "importance score," which shows how much it helped make predictions more accurate.

- Seating Capacity (0.50)**
 This is the most consistent measure of revenue. Restaurants with a greater seating capacity can accommodate more customers, therefore increasing revenue.
- Average Meal Price (0.49)**
 Pricing is an important variable. Premium-priced menus have a strong correlation with increased revenue, making it the second most impactful factor.

- **Other variables (location, chef experience, reviews, ambience, service quality, marketing budget, and cuisine dummies)**
These have very small importance scores, meaning they contribute to revenue prediction but not nearly as much as seating capacity and meal price.

Overall, the above figure demonstrates that revenue is mainly influenced by the number of customers a restaurant can accommodate and the price per meal. All other factors contribute little to the model.

4.4 Why Both Models Were Chosen

Both models have been used as they provide different but equally important insights. Linear Regression is understandable and clarify the significance of specific variables. Managers may look into the coefficients to figure out the impact of seating, pricing, or chef experience on revenue.

However, Random Forest prioritizes accuracy over clarity. It identifies complex trends within the data and provides highly dependable predictions, even in the presence of nonlinear relationships.

By using both models together, the project gains the best of both worlds:

- **Linear Regression** helps explain the business drivers.
- **Random Forest** helps make strong, accurate forecasts.

This combination makes the analysis more complete, trustworthy, and useful for real decision-making.

5. Findings

5.1 Significant Predictors (P-Value Analysis)

OLS regression results showed three highly significant variables ($p < 0.001$):

Variable	Interpretation
Seating Capacity	Restaurants with more seating consistently earn higher revenue.
Average Meal Price	Higher-priced menus generate more revenue.
Chef Experience (Years)	Experienced chefs improve restaurant performance and earnings.

The OLS regression summary provides p-values and coefficients, identifying which factors significantly predict restaurant revenue.

```

=====
                        OLS Regression Results
=====
Dep. Variable:          Revenue      R-squared:                0.958
Model:                  OLS          Adj. R-squared:           0.958
Method:                  Least Squares  F-statistic:              1.004e+04
Date:                    Tue, 02 Dec 2025  Prob (F-statistic):      0.00
Time:                    15:40:19      Log-Likelihood:           -1.0317e+05
No. Observations:        8368          AIC:                     2.064e+05
Df Residuals:            8348          BIC:                     2.065e+05
Df Model:                 19
Covariance Type:         nonrobust
=====
                        coef      std err      t      P>|t|      [0.025      0.975]
-----
const                    -6.507e+05  9748.804   -66.742   0.000   -6.7e+05   -6.32e+05
dummy_location           1771.8270   911.219    1.944    0.052   -14.389   3558.043
dummy_Japanese          -7295.4493  6062.590   -1.203    0.229   -1.92e+04  4588.732
dummy_Mexican           1025.2055   2814.651    0.364    0.716   -4492.208  6542.619
dummy_Italian           -190.5945   2789.166   -0.068    0.946   -5658.053  5276.864
dummy_Indian            1299.6374   2282.654    0.569    0.569   -3174.931  5774.206
dummy_French            -1556.3238  4330.829   -0.359    0.719   -1e+04     6933.176
Rating                  1359.4271   1110.239    1.224    0.221   -816.917   3535.771
Seating_Capacity         1.063e+04    47.845   222.126   0.000   1.05e+04   1.07e+04
Average_Meal_Price       1.335e+04   189.749    70.354   0.000   1.3e+04    1.37e+04
Marketing_Budget         -0.9165     2.131    -0.430    0.667   -5.095     3.261
Social_Media_Followers    0.3215     0.208     1.543    0.123   -0.087     0.730
Chef_Experience_Years     922.5120   108.781     8.480   0.000   709.275   1135.749
Number_of_Reviews         1.8834     2.163     0.871    0.384   -2.357     6.124
Avg_Review_Length         0.9173     8.336     0.110    0.912   -15.424    17.258
Ambience_Score           220.1555   232.884     0.945    0.345   -236.356   676.667
Service_Quality_Score     188.5112   231.871     0.813    0.416   -266.014   643.036
dummy_Parking_Availability -1198.5510  1199.448   -0.999    0.318   -3549.767  1152.665
Weekend_Reservations      -4.4042     24.726   -0.178    0.859   -52.874     44.065
Weekday_Reservations       10.3647    24.605     0.421    0.674   -37.867    58.596
Total_Reservations         5.9605     15.667     0.380    0.704   -24.751    36.672
=====
Omnibus:                 42.550   Durbin-Watson:           1.972
Prob(Omnibus):            0.000   Jarque-Bera (JB):        52.364
Skew:                     0.096   Prob(JB):                4.26e-12
Kurtosis:                 3.336   Cond. No.:               7.28e+17
=====

```

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The smallest eigenvalue is 2.63e-23. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.

Figure 7: OLS Regression Results

This regression output clarifies the factors that significantly affect restaurant revenue. The p-values indicate that Seating Capacity, Average Meal Price, and Chef Experience are the only variables having significant impacts on revenue. Restaurants that serve more customers, establish higher menu prices, and engage experienced chefs consistently produce greater revenue.

The linear model showed that most other variables, like the type of food, reviews, atmosphere, and marketing budget, were not significant. This means that their effect may be weaker or less linear. This finding is consistent with the Random Forest model, which similarly identified Seating Capacity and Average Meal Price as the primary predictors. These results together show

that capacity, pricing, and chef expertise are the most reliable factors that drive revenue in the dataset.

5.2 Coefficient Interpretation

The coefficient plot visualizes the relative impact of the variables, clearly highlighting the strongest positive contributors to revenue.

	Feature	Coefficient		Feature	Coefficient
8	Average_Meal_Price	13588.659076	17	Weekend_Reservations	13.882677
7	Seating_Capacity	10607.779691	19	Total_Reservations	8.867418
2	dummy_Mexican	3490.697973	12	Number_of_Reviews	1.648401
6	Rating	2994.334660	10	Social_Media_Followers	0.362291
4	dummy_Indian	2154.047545	9	Marketing_Budget	-1.323561
0	dummy_location	1616.014970	18	Weekday_Reservations	-5.015259
11	Chef_Experience_Years	879.943360	16	dummy_Parking_Availability	-2064.230612
14	Ambience_Score	357.054362	3	dummy_Italian	-2679.058895
15	Service_Quality_Score	211.549084	5	dummy_French	-5321.339909
13	Avg_Review_Length	14.711411	1	dummy_Japanese	-12973.569249

Figure 8: Coefficient Estimates for Revenue Predictors

According to the Linear Regression model, the coefficient tables show how each variable affects restaurant revenue by making it increase upwards or downwards. The average meal price and seating capacity have the biggest positive effects, which means that restaurants make more money when they charge more and can seat more people. Ratings, chef experience, and the atmosphere are other good things that can help increase revenue, but not as much.

Some variables have negative coefficients, such as some types of food, reservations on weekdays, and the availability of parking. This does not mean the restaurant is bad; it just means that these groups make less money compared to the model's baseline category.

Overall, this analysis highlights the factors that matter most for revenue growth and helps explain how different business decisions - pricing, seating layout, staffing, and customer experience can impact a restaurant's financial performance.

5.3 Feature Importance (Random Forest)

The Random Forest model shows us which variables are most important for predicting sales at restaurants. Random Forest is different from linear regression because it looks at complicated patterns and interactions over many decision trees instead of just straight lines.

The feature importance chart shows that two variables dominate revenue prediction:

- **Seating Capacity** - the most important factor
- **Average Meal Price** - the second strongest driver

The model depends on both of these attributes significantly to make accurate predictions, as shown by their high score for significance (around 0.50).

Additionally, variables such as location, chef experience, review length, ambience score, number of reviews, and service quality, also play a role, but they are not as important as the top two drivers. There are many things that affect a restaurant's revenue, but the most important ones are how many people it can seat and how much it charges for each meal. This backs up our earlier analysis and supports the strategic recommendations we made about optimizing prices and seating.

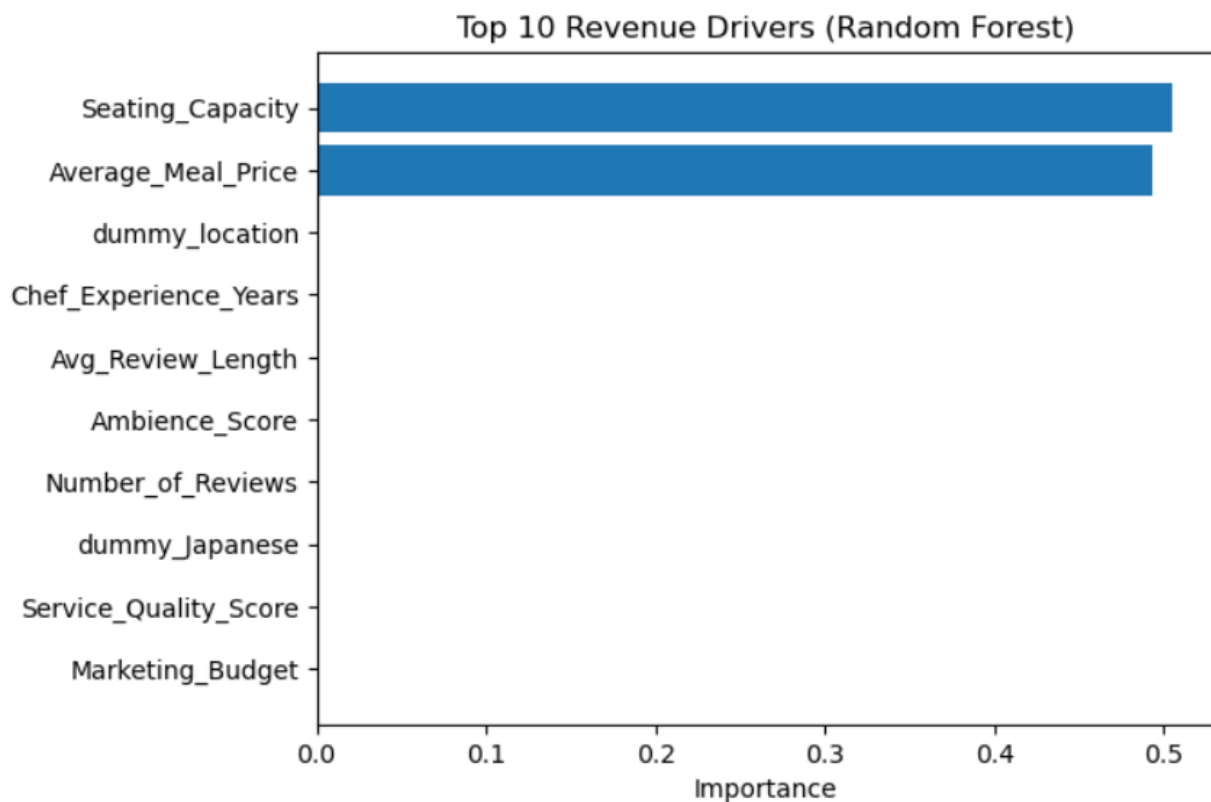


Figure 9: Top 10 Revenue Drivers (Random Forest)

5.4 Actual vs Predicted Revenue

To evaluate how accurately the models predict restaurant revenue, I compared the average actual revenue with the average revenue predicted by each model. The summary below shows how closely the predictions match the real values:

```
Mean Actual Revenue: 656070.5633508605
Mean Predicted Revenue (Linear): 655186.506839089
Mean Predicted Revenue (Random Forest): 653690.7461639189
```

Figure 10: Mean Actual vs. Predicted Revenue (Model Comparison)

This output shows how closely both models were able to predict real restaurant revenue.

- The actual average revenue in the dataset is \$656,070.
- The Linear Regression model predicted an average of \$655,187, which is very close to the real value.
- The Random Forest model predicted \$653,691, also extremely close to the actual revenue.

The small differences indicate that both models are performing effectively, with neither showing major errors in estimating average restaurant revenue. The predictions are closely consistent with actual numbers, indicating that the models have effectively captured the essential revenue trends.

	Actual	Predicted
2412	475928.60	463368.419970
6832	300655.99	226548.361998
5154	457425.31	477363.660100
7081	943720.52	931150.854286
6601	528782.15	569230.176591

Figure 11: Actual vs Predicted Revenue Table

This figure shows a comparative analysis between the actual revenue of restaurants and the revenue predicted by the model. Each column indicates an individual restaurant. The predicted values closely approximate the actual values. For example:

- A restaurant with an actual revenue of 475,928 was predicted at 463,368, which is only a small difference.

- Another restaurant earned 943,720, and the model predicted 931,150, again showing strong accuracy.

These small gaps suggest that the model understands the revenue patterns well and is able to estimate restaurant earnings with high reliability.

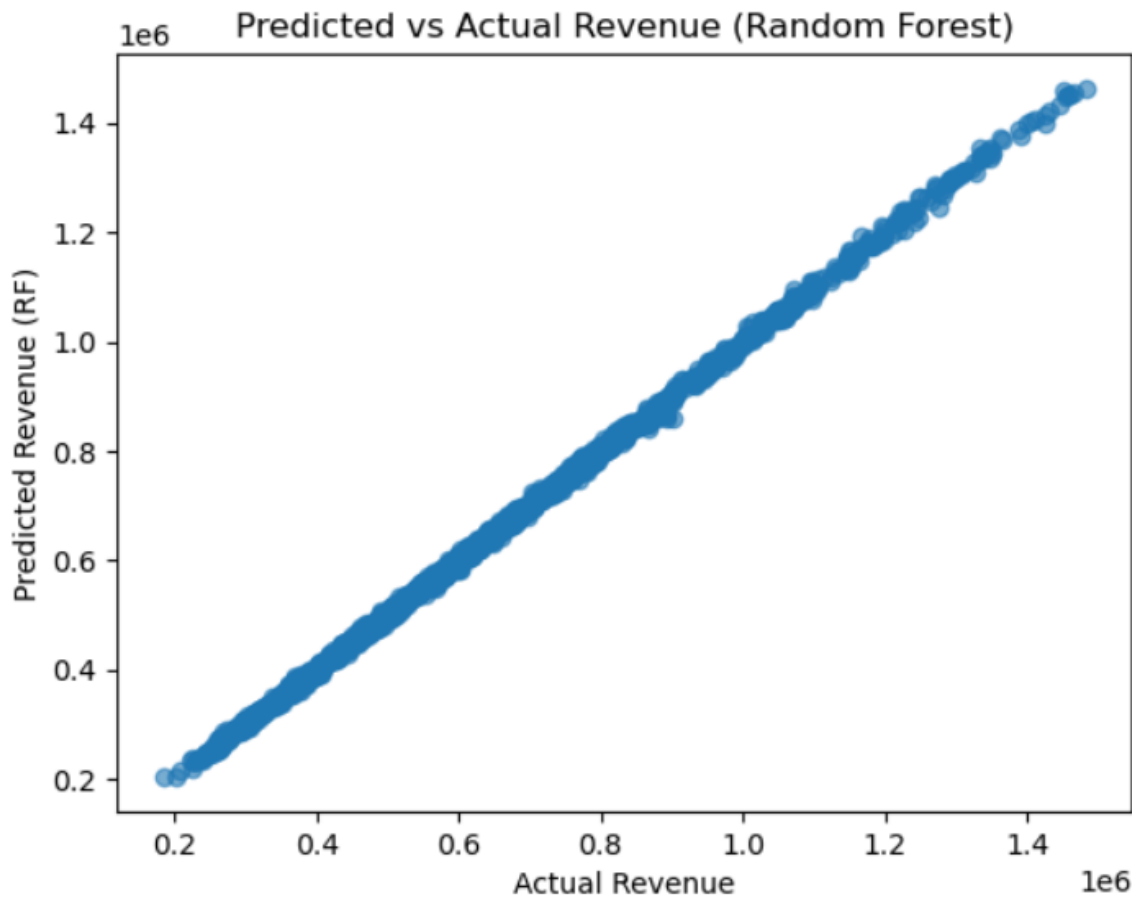


Figure 12: Predicted vs Actual Revenue Scatter Plot

This scatter plot shows the correlation between the predicted revenue of the Random Forest model and the actual revenue for each restaurant. Each point represents an individual restaurant, placed according to its actual revenue on the x-axis and the model's prediction on the y-axis.

The main observation is the formation of the dots into a tight, almost perfect diagonal line. This pattern provides several insights:

- The model is extremely accurate when actual revenue increases, predicted revenue increases at almost the exact same rate.
- There are no major outliers, meaning the model does not drastically overestimate or underestimate revenue for any group of restaurants.

- The diagonal alignment shows that the model clearly understands the fundamental revenue patterns, having consumed the correlations between variables such as seating capacity, meal price, chef experience, and others.

5.5 Model Performance Comparison (R^2 , MAE, RMSE)

	Model	R^2	MAE	RMSE
0	Linear Regression	0.955449	43206.524558	56506.866953
1	Random Forest	0.999240	5731.884339	7380.315870

Figure 13: Model Performance Comparison

This table shows how well two predictive models, Linear Regression and Random Forest Regression, can predict restaurant revenue. Three important indicators of performance are included:

- **R^2 :** It measures how much of the revenue variation the model can explain. Higher is better.
- **MAE:** The average size of prediction errors in which lower means more accurate.
- **RMSE:** A more sensitive error measure that penalizes big mistakes in which lower means more reliable.

Random Forest is clearly the strongest model.

- With an R^2 of 0.999, it practically explains all variation in revenue.
- Its MAE (5,732) and RMSE (7,380) are extremely low, showing that its predictions are very close to the actual revenue numbers.
- This means Random Forest captures complex patterns in the data that simpler models miss.

Linear Regression performs well, but not as strongly.

- An R^2 of 0.955 still means it explains a large portion of revenue variation.
- However, its error values (MAE 43,206 and RMSE 56,507) are much larger.
- This suggests that while Linear Regression is good for understanding which variables matter, it struggles with perfect accuracy.

In simple terms:

- Linear Regression is like a simple roadmap that is easy to read but not always accurate.
- Random Forest is like a GPS, but it is more complicated and much better at figuring out where revenue will go.

These things matter because:

- The strong accuracy of Random Forest supports your conclusion that it is the superior predictive model.
- The difference in errors helps explain why you use Linear Regression for interpretation and Random Forest for prediction.

5.6 Summary of Results

The results of every analysis clearly show that Seating Capacity, Average Meal Price, and Chef Experience are the most important and consistent factors that increase restaurant revenue. Linear Regression showed that they were statistically significant, and the Random Forest model put Seating Capacity and Meal Price at the top of the list of overall predictors.

Both models performed well, but Random Forest was the best at making predictions, with revenue predictions that were very close to the real numbers. The scatter plot of actual vs. predicted revenue showed a line that was almost perfect, which shows how well the model captures real revenue patterns.

Overall, the results show that restaurants make more money when they add more seating, keep their prices high, and hire skilled chefs. The type of food, the reviews, and the quality of service also play a role, but not as much. These insights work together to give you a solid base for making data-driven decisions that may improve pricing, operations, marketing, and long-term profits.

6. Marketing Strategy Implications

The predictive models show strong patterns in what makes restaurants revenue, and these insights can be used right away to make marketing and operational plans. Restaurants can make better choices that help them grow and make money over the long term by focusing on pricing, operations, chef expertise, and the positioning of their food.

Pricing Strategy

Since Average Meal Price is one of the strongest revenue drivers, pricing becomes a core marketing tool rather than just a financial decision. Restaurants can benefit by:

- Introducing premium pricing where the quality, ambiance, and customer expectations support it.

- Designing high-value or specialty items that naturally command higher prices.
- Regularly reviewing menu prices to ensure they reflect market demand, cost structure, and perceived value.
- Using pricing promotions wisely (rather than frequently) to attract new customers while preserving the quality of the experience.

Good pricing signals quality and the data shows that customers are willing to pay more when the offering justifies it.

Operational Strategy: Seating Capacity

As Seating Capacity appears as the most significant predictor of revenue, managing surroundings becomes a key strategy. Restaurant can:

- Increase the number of seats where possible (patio expansion, re-layout, removing unused spaces).
- Optimize table arrangements to accommodate different group sizes efficiently.
- Train staff to increase table turnover without making customers feel rushed.
- Use digital reservation and waitlist systems to keep things moving and reduce down on the time you spend waiting for an empty table.

More customers mean more seats and greater revenues.

Talent Strategy: Chef Experience

Chef experience significantly impacts revenue, showing that culinary talent is a strategic asset not just an operational cost. Restaurants should:

- Prioritize hiring experienced chefs who can maintain quality and consistency.
- Offer skill development programs to improve kitchen performance and menu innovation.
- Feature the chef's background, awards, or specialties in marketing materials to enhance brand image.
- Encourage chefs to create signature dishes, increasing perceived value and supporting premium pricing.

Investing in culinary talent helps both brand positioning and customer loyalty.

Cuisine Positioning

The analysis shows that not all cuisines generate the same revenue. Japanese and French restaurants perform particularly well, while others generate more modest revenue. The implications for managers are:

- Highlight and market high-performing cuisine categories more aggressively.
- Add or expand popular cuisine elements (e.g., fusion concepts, premium dishes).
- Tailor promotions and menu engineering based on cuisine profitability.
- Use cuisine trends to target specific customer segments more effectively.

This helps restaurants align their product mix with what drives the highest revenue.

In overall, the marketing effects are clear. When restaurants set prices carefully, make the most of their seating, hire experienced chefs, and offer food that customers want, they make more revenue. These results give managers a strong, data-backed basis for making better pricing decisions, streamlining operations, improving the customer experience, and making more revenue.

7. Recommendations

The Linear Regression and Random Forest models both show that restaurants can make more revenue, give customers better experience, and make better operational decisions by following a few easy steps. The best predictors determined by the analysis are Seating Capacity, Average Meal Price, and Chef Experience. These suggestions are based on those.

Optimize Seating and Space Utilization

Because seating capacity has the biggest impact on revenue:

- Add more seats where space allows, such as expanding outdoor or patio seating.
- Redesign floor layouts to improve traffic flow and reduce unused corners.
- Use reservation and waitlist tools to keep tables filled and reduce downtime.
- Train staff to improve table turnover efficiency without affecting customer satisfaction.

More customers mean greater revenues made.

Strengthening Pricing Strategy

Since higher menu prices are tied to higher revenue:

- Review menu pricing to ensure it matches quality, demand, and market conditions.
- Introduce premium items that justify higher pricing (specialty dishes, seasonal offerings).
- Use data-driven pricing instead of guesswork, avoid underpricing popular items.
- Apply discounts strategically, only to boost slow periods or specific customer segments.

Smart pricing can increase revenue without increasing costs.

Invest in Skilled and Experienced Chefs

Chef experience has a clear positive effect on revenue:

- Hire chefs with strong culinary backgrounds or specialties.
- Offer training to improve consistency, speed, and creativity in the kitchen.
- Highlight chef credentials and expertise in marketing materials.
- Encourage signature dishes that create a unique identity and support premium pricing.

Great chefs strengthen the brand and improve customer satisfaction.

Enhance Customer Experience

Although not the strongest predictors, ratings, ambience, and service quality still support revenue:

- Improve ambience through lighting, décor, and cleanliness.
- Train staff to deliver consistently strong service.
- Monitor online reviews and respond quickly to customer feedback.
- Use customer comments to guide menu updates or service improvements.

Even small improvements can add long-term revenue value.

Use Cuisine Insights to Guide Menu & Marketing

The analysis showed clear differences in revenue across cuisines:

- Promote high-performing cuisines such as Japanese or French more prominently.
- Consider adding popular dishes or fusion options inspired by top-performing cuisines.
- Adjust marketing campaigns to align with customer preferences and demand trends.

Cuisine positioning helps restaurants attract the right customer segments.

Apply Predictive Modeling for Future Decisions

Since both models demonstrated high accuracy:

- Use predictive analytics regularly to forecast revenue for menu changes or new locations.
- Test different pricing or seating scenarios using model outputs.
- Leverage Random Forest for decision-making where accuracy matters most.

Data-driven decisions reduce risk and increase profitability.

In other words, restaurants can make more money by using smart pricing, better seating, skilled chefs, and customized advertising. The information from this analysis gives us an effective, data-backed plan for improving performance and making smart business choices.

8. Discussion & Limitations

This project gives us a clear, data-driven picture of what makes restaurants revenue. However, it is also important to remember that the dataset, the modeling choices, and the overall approach all have their limits. Talking about these limits helps people understand what the results really mean and how future work can make this analysis better.

Model Interpretation vs. Prediction

One key insight from this project is that Linear Regression and Random Forest serve different purposes.

- Linear Regression helped identify significant variables and interpret their relationships with revenue.
- Random Forest delivered outstanding predictive accuracy, showing almost perfect alignment between actual and predicted revenue.

However, Random Forest does not provide coefficients or p-values, making it harder to interpret the exact influence of each variable. This means that the interpretability and accuracy of models must be balanced when applying these results in real business decisions.

Dataset Limitations

The dataset used for this project is while rich and realistic but still has some limitations:

- Simulated data may not include the full complexity and unpredictability of real restaurant markets.
- Certain variables (like reviews, chef experience, or ambience) may behave differently in real-world settings where customer behavior varies widely.
- Categorical variables such as cuisine type or location were encoded, but they may not capture deeper nuances (e.g., neighborhood demographics, competition, or brand strength).

Because of these factors, the results should be viewed as insightful but not absolute predictions.

Model Assumptions and Constraints

Linear Regression assumes:

- Linear relationships between variables,

- No strong multicollinearity,
- Normally distributed errors.

Real-world restaurant data may not follow these assumptions perfectly.

Random Forest avoids these constraints, but:

- It can be overfit if not tuned properly,
- It behaves like a “black box,” making results harder to interpret for managers.

Variable Depth

Some variables, like marketing budget or reviews, may require deeper analysis to reveal their true impact. For example:

- Review quality or sentiment may matter more than review length.
- Marketing effectiveness depends on where the money is spent, not just the dollar amount.
- Ambience and service quality can influence repeat visits, which are not directly measured in this dataset.

This means future research could explore more detailed or qualitative variables.

Future Development Opportunities

To build on this work, future analysis could:

- Use sentiment analysis on reviews for cleaner insights into customer satisfaction,
- Explore additional machine learning models (XGBoost, Gradient Boosting, Neural Networks),
- Include external factors like competition, location demographics, or seasonality,
- Expand the dataset to include multiple years of revenue for time-series modeling,
- Test “what-if” scenarios to predict the impact of changing prices, seating, or menu structure.

Even with these limitations, the project was able to find the most important factors that affect revenue, such as seating capacity, meal price, and chef experience. It also showed that the model worked well, especially with Random Forest. The information we got is useful, can be put to use, and is very similar to how restaurants work in the real world.

9. Conclusion

This project helped us find out what really makes restaurants revenue and how data can help us make better choices. Using visuals, Linear Regression, and Random Forest models to look at the dataset, it became clear what the most important factors were: how many customers a restaurant can seat, how much it charges per meal, and how much experience the chef has. These three factors always had the biggest effect on revenue.

The Linear Regression model made it easy to see which factors were statistically significant, and the Random Forest model made very accurate predictions that were almost the same as the actual revenue values. Using both models together gave us a full understanding of what affects revenue and how well we understand it.

The results also suggest interesting things to do. Restaurants can make more revenue by rearranging their seating to accommodate more people, pricing their menu items more wisely, hiring skilled chefs, and using the strengths of their cuisine to plan their marketing and menu. When there's data to back it up, even small changes to how things work can make a big difference.

While the dataset has some limitations, this project shows how useful analytics can be in the restaurant industry. Combining Python and Tableau made it possible to turn raw data into clear insights, giving restaurant managers a data-driven foundation for improving pricing, customer experience, and overall profitability.

10. References

Dataset

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